Vision-Language-Action Navigation for a Quadruped Robot

Project Description

This project focuses on developing a vision-language-based autonomous navigation framework for the Unitree Go2 quadruped robot. The first stage of the project will establish the required robotic infrastructure by integrating the Go2 with an NVIDIA Jetson Orin, accessing the robot's available sensors, and sending high-level control commands. A corresponding simulation environment will also be developed using a platform such as NVIDIA Isaac Sim, allowing the same sensing and high-level control pipeline to be tested safely before deployment on the physical robot.

The main objective is to integrate a pretrained Vision-Language-Action (VLA) model that enables the robot to interpret visual observations together with natural-language instructions and generate appropriate navigation actions. The system will first be implemented and evaluated in simulation and subsequently transferred to the real Go2 robot. Depending on project progress, additional components can include Vision-Language-Model-based navigation with classical planning, semantic mapping of the environment, and language-conditioned object-goal navigation.

Main Tasks

Task 1: NVIDIA Orin–Go2 Integration and High-Level Control

Set up communication between the NVIDIA Jetson Orin and the Unitree Go2, access the available robot sensors and states, and send high-level commands such as forward and backward motion, turning, velocity control, and stopping.

Task 2: Go2 Simulation Environment

Set up the Unitree Go2 in a robotic simulator such as NVIDIA Isaac Sim. The student should be able to access the simulated sensors and control the simulated robot using high-level actions similar to those used for the real Go2.

Task 3: VLA-Based Control in Simulation

Integrate or adapt a pretrained Vision-Language-Action model in the simulation environment. The simulated Go2 should receive visual observations together with natural-language instructions and generate appropriate high-level navigation actions.

Task 4: VLA Deployment on the Real Go2

Deploy the VLA pipeline on the physical Unitree Go2 using the NVIDIA Jetson Orin. Evaluate the ability of the system to understand language-conditioned navigation tasks and execute the corresponding actions in real-world environments.

Additional Tasks

Additional Task 1: VLM + Classical Navigation

Use a Vision-Language Model (VLM) to understand the visual scene and natural-language instruction, identify or determine the desired navigation target, and pass this target to a conventional

navigation and path-planning system for execution.

Additional Task 2: Semantic Mapping with Go2

Develop a semantic mapping system for the Go2 that combines geometric information about the environment with semantic information such as chairs, doors, tables, hallways, stairs, and other landmarks. The resulting map can support higher-level and language-based navigation tasks.

Additional Task 3: Language-Conditioned Object-Goal Navigation

Develop a system in which the Go2 receives natural-language commands such as "find the chair" or "go to the backpack," identifies or grounds the requested object using visual perception, and autonomously navigates toward the corresponding target.